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Direct simulations of external flow and noise radiation using the generalized interpolation-supplemented cascaded lattice Boltzmann method

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Highlights

- The cascaded collision model is implemented with the body-fitted mesh.
- A novel index algorithm is proposed to reduce memory usage by implicit streaming.
- Cases are validated for noise radiation with O-type and C-type meshes.

Abstract

Direct simulations of external flow and the associated noise radiation are studied by an improved lattice Boltzmann method, i.e., the generalized interpolation-supplemented cascaded lattice Boltzmann method (GICLBM). In this method, the cascaded collision scheme is used to improve the numerical stability of the conventional collision schemes, and the generalized interpolation approach is used in the particle streaming process so as to allow a non-uniform and body-fitted mesh partition. With that, both near- and far-field flow dynamics and noise radiation are resolved simultaneously. In order to capture sound waves, the perfectly matched layer is also implemented so as to avoid waves reflecting to and polluting the inner acoustic field. Moreover, a novel index technique is developed for the GICLBM to enable implicit streaming, which brings an efficient memory reduction. Three cases are then performed to showcase the feasibility, accuracy, extensibility, and efficiency of the present framework, including flow past a square cylinder, flow past an elliptic cylinder, and flow past a NACA 0012 airfoil, each implemented with a type of body-fitted mesh. Both the fluid dynamic and noise radiation are found to be in good agreement with results using the Navier–Stokes solvers. This study demonstrates the potential of the GICLBM for accurately and efficiently simulating external problems as well as sound generation and propagation.

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Keywords

Lattice Boltzmann method; Flow induced noise; Body-fitted mesh; External flow

1. Introduction

Flow-induced noise widely exists in nature and engineering applications, such as aircraft, automobiles, and underwater vehicles. Among them, the noise generated from the flow past cylinders with various complex cross-section shapes, such as triangles [1], squares [2], [3] and circles [4], has arouse strong interests. The flow separates from the surface of the structure and the appearance of the alternately vortex shedding will radiate intense noise where tones are dominant. A comprehensive understanding of sound generation and propagation is an essential basis for noise reduction.

Direct acoustic simulation is an important tool in computational aeroacoustic (CAA). It is usually utilized for investigating the acoustic mechanism because no additional hypothesis is introduced. Lattice Boltzmann method (LBM) is confirmed a low-dispersion and low-dissipation approach, especially for the acoustic waves' propagation through the von Neumann analysis [5], [6]. And as collision models in the LBM developed, the computational stability has been significantly enhanced. Therefore, the LBM is considered a promising numerical method in CAA [7], [8], [9]. However, there are still some obstacles to the LBM implemented in acoustic problems. Firstly, the sound waves propagate at the speed of sound, but hydrodynamic waves only move at the background velocity. The scale of the sound wave is much larger than that of the vortex structure, so the mesh needed for the far-field noise is much coarser compared with the mesh adjacent to the structure. The LBM in the uniform mesh will waste a huge amount of memory. As for the widely used multi-block method, it is found that significant spurious noise will occur when the vortex crosses the interface between fine and coarse meshes [10]. Astoul et al. [11] proposed an "aeroacoustically-compliant" algorithm to eliminate this spurious noise without the overlapping region, but the computation complexity also increased. In addition, Bellotti et al. [12] developed an adaptive multi-resolution approach that can guarantee accuracy by error control with wavelet transform. This approach has been successfully applied to hyperbolic equations. Besides, another concern is the curvilinear boundary reconstructed by the uniform mesh through classical methods [13], [14], [15] should be reconsidered for CAA problems. The stair-shaped boundary with the insufficient mesh may scatter false noise [16]. This problem should be noticed especially when the acoustic wavelength is close to the structure.

The generalized interpolation-supplemented lattice Boltzmann method (GILBM) [17] calculated on a non-uniform body-fitted mesh is a feasible way to handle the above problems. This method originated from the interpolation-supplemented lattice Boltzmann method (ISLBM) which was proposed by He and Doolen [18]. However, the mesh in the ISLBM is generated algebraically and the coordinate transformation is controlled by the analytic derivative. Imamura et al. [17] developed the ISLBM to an arbitrarily generated non-uniform mesh called GILBM. The streaming process of distribution functions between different lattices is dealt with the interpolation, and the computational stability is satisfied to the Courant–Friedrichs–Lewy (CFL) condition. Since only the streaming process is modified, the advanced collision models such as the multi-relaxation-time model [19], the cascaded collision model [20], [21] and the regularized model [22] can be directly implemented to the GILBM to improve the numerical stability. Since the GILBM has a flexible mesh configuration, some successful applications in the flow past a bluff body [23] and heat transfer problems [24], [25], [26] have been reported, but its performance in acoustic problems has not yet to be discussed. Moreover, the non-reflective buffer layer is usually utilized in the direct acoustic simulation to dissipate the sound waves reflected from the computational boundary [27]. The mesh resolution is relatively high in the near-wall region to resolve the flow structure. And the approximate mesh resolution is required in the region where sound waves linearly propagate. However, the mesh resolution in the buffer layer is much coarser. In fact, this mesh configuration can be easily implemented for the GILBM.

In acoustic problems, the concerned region is usually far away from the sound sources, and the computational domain required for the direct acoustic simulation is vast. Even for the GILBM, where the amount of meshes can be significantly reduced, an enormous memory consumption is still required. Therefore, there is a strong demand for reducing the memory usage. Similar to the classical LBM, the GILBM requires two sets of meshes to implement reading and writing operations during the streaming process. In the LBM, there are many interesting and efficient algorithms for reducing memory usage. Bailey et al. [28] have proposed A–A pattern that two streaming and collision processes are required for even and odd time steps, respectively. For simplifying the above processes, Mohrhard et al. [29] developed a shift and swap streaming (SSS) algorithm. By swapping and shifting pointers, only one streaming and collision process is satisfied. More recently, Kummerländer et al. [30] and Ma et al. [31] further developed implicit streaming algorithms to improve the utilization of the bandwidth, and called the periodic shift (PS) and the one-step index (OSI) algorithm, respectively. Namvar and Leclaire [32] discussed several key issues for optimizing LBM memory usage and proposed an algorithm to significantly reduce memory usage by rearranging the memory layout. However, due to the post-streaming distribution functions being interpolated from the adjacent lattices, additional data dependencies are in existence for the GILBM. The one mesh algorithm developed for the GILBM is still in the absence of the author's knowledge. In the present work, a low-memory usage algorithm for the GILBM is proposed, where only one mesh with an extra distribution function array is required to store post-streaming distribution functions. The implicit streaming process could be efficiently executed by calculating the index of distribution functions in the memory address.

This paper intends to explore the feasibility of the GILBM in CAA problems and propose a streaming algorithm to reduce the memory usage of the GILBM. The cascaded collision model is implemented to improve computational stability. Therefore, the algorithm being introduced and discussed is named the generalized interpolation-supplemented cascaded lattice Boltzmann method (GICLBM). Moreover, the perfectly matched layer (PML) technology is utilized to eliminate the noise reflected from the outer boundary [27] that could pollute the inner acoustic field. The flow and noise generated from a square cylinder, an elliptic cylinder, and a NACA 0012 airfoil are considered, so as to evaluate the accuracy of the GICLBM in dealing with flow-induced noise problems.

2. Methodology

2.1. The generalized interpolation-supplemented cascaded lattice Boltzmann method

The lattice Boltzmann method stems from the kinetic theory of gases at the mesoscopic scale, and is developed for solving weakly compressible Navier–Stokes equations. In the $DdQq$ scheme [33], where d denotes the spatial dimension and q is the quantity of discrete velocities $\boldsymbol{e_i}$, the distribution functions f_i relax to the local equilibrium f_i^{eq} in the collision process, and stream to adjacent lattices. He and Doolen [18] extended the LBM from a uniform orthogonal mesh to a body-fitted mesh based on the coordinate transformation. After that, Imamura et al. [17] demonstrated a more generalized form for transforming from the physical domain (x, y), to the computational domain (ξ, η). To improve the numerical stability, the cascaded particle collision model [21], [34], [35] is implemented in the GILBM. The GICLBM without the body force term also follows the general two-step formulation of the LBM, and can be expressed as

$$\begin{aligned} &\textbf{collision step:} \quad f_i^{*}(\boldsymbol{\xi}, t) = \boldsymbol{M}^{-1}\boldsymbol{N}^{-1}[(\boldsymbol{I}-\boldsymbol{\Lambda})T_j(\boldsymbol{\xi}, t) + \boldsymbol{\Lambda} T_j^{eq}(\boldsymbol{\xi}, t)], \\ &\textbf{streaming step:} \quad f_i(\boldsymbol{\xi}, t+\Delta t) = f_i^{*}(\boldsymbol{\xi_i} - \Delta\boldsymbol{\xi_i}, t), \end{aligned} \tag{1}$$

where $f_i^{*}(\xi, t)$ is the post-collision distribution function at the time t and the lattice position $\boldsymbol{\xi}$ in the computational domain and $\boldsymbol{I}$ is an identity matrix. The macroscopic quantities can be calculated as

$$\rho = \sum_{i=0}^{8} f_i, \quad \rho\boldsymbol{u} = \sum_{i=0}^{8} \boldsymbol{e_i} f_i, \quad i = 0, 1, \ldots, 8, \tag{2}$$

where the density ρ and the velocity vector $\boldsymbol{u} = [u_x, u_y]^{\top}$ can be calculated by the Einstein summation convention. The pressure is satisfied $p = (\rho - \rho_0)c_s^2$ in the isothermal condition. In general, the sound speed is $c_s = 1/\sqrt{3}$ [33] and the reference fluid density ρ_0 is set as $\rho_0 = 1$. The $D2Q9$ scheme is used for present two-dimensional problems.

In the cascaded particle collision model, T_i represents the combination of the central moment of distribution function, which can be calculated as [35]

$$\boldsymbol{T} = [k_{00}, k_{10}, k_{01}, k_{20} + k_{02}, k_{20} - k_{02}, k_{11}, k_{21}, k_{12}, k_{22}]^{\top}, \tag{3}$$

where the central moment k_{mn} is defined as

$$k_{mn} = \sum_{i=0}^{8} f_i (e_{ix} - u_x)^m (e_{iy} - u_y)^n. \tag{4}$$

The relationship between the multi-relaxation-time model and the cascaded particle collision model is clarified by Fei and Luo [35]. The distribution function $\boldsymbol{f_i}$ is firstly transformed to the moment space with the transformation matrix $\boldsymbol{M}$, and is then converted to the central moment space with the shift matrix $\boldsymbol{N}$. This operation can be expressed as $\boldsymbol{T} = \boldsymbol{NMf}$. More information about this model and the details of $\boldsymbol{N}$ and $\boldsymbol{M}$ can be found from Ref. [35]. Moreover, the matrix formulation is also generalized for rectangular [36] and cuboid lattices [37]. The corresponding equilibrium T_i^{eq} is calculated by the continuous Maxwell–Boltzmann distribution in the velocity space $\boldsymbol{e} = [e_x, e_y]^{\top}$ [20], and can be simplified as

$$T_i^{eq} = \rho\left[1, 0, 0, 2c_s^2, 0, 0, 0, 0, 2c_s^4\right]^{\top}. \tag{5}$$

$\boldsymbol{\Lambda}$ is the relaxation matrix, which is expressed as

$$\boldsymbol{\Lambda} = diag\left(s_0, s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8\right), \tag{6}$$

which should satisfy $s_1 = s_2$, $s_4 = s_5$ and $s_6 = s_7$ [35].

In the GICLBM, the collision step is performed in a fully localized manner, but the streaming step needs to read information from adjacent lattice nodes. The coordinate transformation is the core part of the GICLBM for calculating on body-fitted meshes, which basically satisfies

$$\begin{bmatrix} \partial_x \xi & \partial_y \xi \\ \partial_x \eta & \partial_y \eta \end{bmatrix} = \begin{bmatrix} \partial_\eta y & -\partial_\eta x \\ -\partial_\xi y & \partial_\xi x \end{bmatrix} J^{-1}, \tag{7}$$

where the Jacobian is calculated by $J = \partial_\xi x \partial_\eta y - \partial_\eta x \partial_\xi y$. Due to the distribution functions being unable to directly stream from one lattice node to another on the uniform mesh in one time step, the contravariant velocity $\widetilde{\boldsymbol{e_i}}$ in the physical domain is used to replace $\boldsymbol{e_i}$ in the computational domain. This velocity $\widetilde{\boldsymbol{e_i}}$ is determined by the non-uniform mesh in the physical domain, and is given as

$$(\tilde{e}_{i,1}, \tilde{e}_{i,2}) = (e_{i,1}, e_{i,2}) \begin{bmatrix} \partial_x \xi & \partial_x \eta \\ \partial_y \xi & \partial_y \eta \end{bmatrix}. \tag{8}$$

As suggested by Imamura et al. [17], the streaming distance in the computational domain can be calculated by the two-step Runge–Kutta (RK2) scheme, i.e.,

$$\begin{aligned} &\text{step I:} \quad \Delta\boldsymbol{\xi}_i^{(1)} = \tfrac{1}{2}\Delta t \widetilde{\boldsymbol{e_i}}\left(\boldsymbol{\xi}\right), \\ &\text{step II:} \quad \Delta\boldsymbol{\xi_i} = \Delta t \widetilde{\boldsymbol{e_i}}\left(\boldsymbol{\xi_i} - \Delta\boldsymbol{\xi}_i^{(1)}\right) + O\left(\Delta t^3\right), \end{aligned} \tag{9}$$

where both $\widetilde{\boldsymbol{e_i}}\left(\boldsymbol{\xi} - \Delta\boldsymbol{\xi}_i^{(1)}\right)$ and $f_i{}^*\left(\boldsymbol{\xi} - \Delta\boldsymbol{\xi_i}, t\right)$ are obtained by the second-order interpolation scheme, which will require data from more adjacent mesh nodes than the classical LBM. As suggested by He and Doolen [18], the second-order upwind quadratic interpolation is utilized to reduce the numerical dissipation, and given as

$$\phi_i\left(\boldsymbol{\xi_i} - \Delta\boldsymbol{\xi_i}\right) = \sum_{k=0}^{2}\sum_{l=0}^{2} a_{i,k,2} a_{i,l,1} \phi_i\left(\xi + k \cdot i_d, \eta + l \cdot j_d\right), \tag{10}$$

where the subscripts i_d and j_d are the upwind directions, respectively. The interpolation coefficients are defined as

$$a_{i,0,\beta} = \frac{1}{2}\left(\left|\Delta\boldsymbol{\xi}_{i,\beta}\right| - 1\right)\left(\left|\Delta\boldsymbol{\xi}_{i,\beta}\right| - 2\right), \quad (11)$$
$$a_{i,1,\beta} = -\left|\Delta\boldsymbol{\xi}_{i,\beta}\right|\left(\left|\Delta\boldsymbol{\xi}_{i,\beta}\right| - 2\right),$$
$$a_{i,2,\beta} = \frac{1}{2}\left|\Delta\boldsymbol{\xi}_{i,\beta}\right|\left(\left|\Delta\boldsymbol{\xi}_{i,\beta}\right| - 1\right),$$

where the subscripts $\beta = 1, 2$ denote the ξ and η direction, respectively. As for the nodes near the boundary, the interpolation functions are given as

$$\phi_i\left(\boldsymbol{\xi}_i - \Delta\boldsymbol{\xi}_i\right) = \sum_{k=-1}^{1}\sum_{l=-1}^{1} a_{i,k,2}a_{i,l,1}\phi_i\left(\xi + k, \eta + l\right), \quad (12)$$

where the interpolation coefficients are

$$a_{i,-1,\beta} = \frac{1}{2}\Delta\boldsymbol{\xi}_{i,\beta}\left(\Delta\boldsymbol{\xi}_{i,\beta} - 1\right), \quad (13)$$
$$a_{i,0,\beta} = 1 - \Delta\boldsymbol{\xi}_{i,\beta}{}^2,$$
$$a_{i,1,\beta} = \frac{1}{2}\Delta\boldsymbol{\xi}_{i,\beta}\left(\Delta\boldsymbol{\xi}_{i,\beta} + 1\right).$$

Due to the calculation of the contravariant velocity and its integration being expensive, the streaming distance $\Delta\boldsymbol{\xi}_i$ is usually calculated at the initialization and written in an extra array.

The time step size Δt is limited by the CFL condition,

$$\Delta t = \lambda \min_{i,\beta,\xi,\eta}\left|\frac{1}{\tilde{e}_{i,\beta}|_{\xi,\eta}}\right|, \quad (14)$$

where the CFL number λ in the present work is fixed at 0.9, which satisfies the stability condition, $\lambda \in (0, 1]$. It should be noted that the shear viscosity ν and the bulk viscosity ζ will also change with Δt,

$$\nu = c_s^2\left(\frac{1}{s_4} - \frac{1}{2}\right)\Delta t, \quad \zeta = c_s^2\left(\frac{1}{s_3} - \frac{1}{2}\right)\Delta t. \quad (15)$$

In other words, s_4 is determined by both the shear viscosity ν and Δt.

2.2. Non-reflective far-field buffer layer

Eliminating the noise reflected from the computational boundary is quite essential in acoustic problems. The acoustic results are generally in a small order and could be easily polluted by reflected spurious sound waves. Najafi-Yazdi and Mongeau [27] proposed a non-reflective boundary condition for the LBM based on the idea of the PML [38]. The term $\Omega_i{}^{PML}$ can be considered as an extra term in the collision process which is formulated as

$$f_i{}^*\left(\boldsymbol{\xi}, t\right) = \boldsymbol{M}^{-1}\boldsymbol{N}^{-1}\left[(\boldsymbol{I} - \boldsymbol{\Lambda})T_j\left(\boldsymbol{\xi}, t\right) + \boldsymbol{\Lambda}T_j{}^{eq}\left(\boldsymbol{\xi}, t\right)\right] + \Omega_i{}^{PML}, \quad (16)$$

where

$$\Omega_i{}^{PML} = -\sigma\left[\boldsymbol{e}_i \cdot \nabla Q_i + 2\left(f_i^{eq}(\rho, \boldsymbol{u}) - f_i^{eq}(\rho_0, \boldsymbol{u}_0)\right) + \sigma Q_i\right], \quad (17)$$

and the additional term Q_i is defined as

$$\frac{\partial Q_i}{\partial t} = f_i^{eq}(\rho, \boldsymbol{u}) - f_i^{eq}(\rho_0, \boldsymbol{u}_0). \quad (18)$$

Here f_i^{eq} denotes the equilibrium distribution function and is expressed as

$$f_i^{eq} = w_i\rho\left[1 + \frac{\boldsymbol{e}_i \cdot \boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{e}_i \cdot \boldsymbol{u})^2}{2c_s^4} - \frac{\boldsymbol{u}^2}{2c_s^2}\right]. \quad (19)$$

Eq. (18) can be easily solved with the second-order finite difference scheme for the spatial gradient, but the computational efficiency will be worse off due to the information reading from adjacent lattices. The performance of the PML is also affected by the damping coefficient σ. As suggested by Xu and Sagaut [39], the profile for σ varies from zero to the maximum value σ_0, and is expressed as

$$\sigma(\boldsymbol{x}) = \sigma_0 \frac{3125(L_{\mathrm{III}} - \boldsymbol{x} + \boldsymbol{x_0})(\boldsymbol{x} - \boldsymbol{x_0})^4}{256 L_{\mathrm{III}}^5}, \tag{20}$$

where L_{III} is the width of the non-reflective buffer layer and $\boldsymbol{x_0}$ is the interface position. The present σ_0 is set to 0.001 to avoid the adverse effects caused by larger damping coefficients [27]. Moreover, the width of the PML should be larger than the wavelength of a single sound wave, and coarse meshes are adopted in this region.

In the present work, the no-slip and no-penetration wall boundary condition and the far-field Dirichlet boundary condition are realized using the non-equilibrium extrapolation scheme [40].

2.3. A low-memory usage streaming algorithm for the GICLBM

In the original GILBM [17], there are two independent steps including the streaming and collision, which are similar to the LBM. As for the LBM, the streaming step is in essence the index of distribution functions in memory changes. Hence, many streaming algorithms have appeared in the LBM to improve the computational efficiency and reduce the memory usage [28], [29], [30], [30]. However, the distribution functions cannot directly stream between lattices for the non-uniform mesh in the GILBM, rather the interpolation and the coordinate transformation. The distribution functions after streaming are determined by several adjacent mesh points. Since a single distribution function is read and written in strict order, it is hard to operate distribution functions as flexibly as the LBM. Noticing that the streaming step of the distribution function is still independent of each other, a one-mesh algorithm by implicitly streaming for the GICLBM is proposed to reduce the memory consumption. This algorithm is quite simple and can be directly implemented with very few changes.

In $DdQq$ schemes, the distribution functions with q directions are stored as the array of structure (AOS). The index in original algorithm can be calculated by the subscript,

$$f_{\xi,\eta,i}\,(i) = f[q \times (\eta \times NX_\xi + \xi) + i]. \tag{21}$$

It is usually necessary to allocate memory for another array with the same size of f to receive post-streaming distribution functions. But in the present work, we have observed that only $q+1$ distribution functions are enough for a single mesh point. The post-collision distribution function will be streaming to the next direction, and the corresponding index of the distribution function in the next time level can be easily calculated by the total computation time t. The key idea of this algorithm is that distribution functions in different directions are independent of each other. Fig. 1 depicts how the indexes of distribution functions in the array change on a single point for the first time step. In the i direction, the distribution function f_{i+1} is given by interpolating f_i^* from adjacent lattice nodes. The distribution function located at the end of the array will be shifted to the head by $(i+t+1)\%q$. The Eq. (21) for the original GICLBM can be developed to

$$f_{\xi,\eta,i} = f[(q+1) \times (\eta \times NX_\xi + \xi) + (i+t)\%q]. \tag{22}$$

Taking the $D2Q9$ scheme for example, the pseudo-code of collision and streaming steps is outlined in Algorithm 1. It should be noted that the order of the code execution along different directions is opposite to the assigned direction. If one wants to get the post-streaming f_8, the first procedure is to read the data of post-collision f_8^* needed for interpolation from the memory address at the current time step. For example, $f_8^*(0,0)$ is located at $f_{\xi,\eta}[(8 + \text{Time Step})\,\%10]$ as shown in Fig. 2. Then the interpolation operation is conducted with Eqs. (10), (12), whose results are then written to $f_{\xi,\eta}[(8 + \text{Time Step} + 1)\,\%10]$. After the streaming step of f_8 for all lattice nodes is completed, the streaming step of f_7 follows. Because the streaming step always ends at the distribution function that does not participate in the collision step, distribution functions along other directions are already read before being overwritten. Since the present algorithm is carried out for a single mesh node in a fully localized manner, it can be easily implemented to all mesh nodes without affecting the overall parallel efficiency.

Algorithm 1 Pseudo-code of collision and streaming procedure in the GICLBM at t with the $D2Q9$ scheme

Collision

for each $i \in [0,9)$ **do**
  $id = 10 \times (\eta \times NX_{\xi} + \xi)$
  $f_i = f[id + (i+t)\%10]$
end for
Update Q by Eq. (18) and calculate Ω^{PML} by Eq. (17) for the non-reflective far-field buffer layer
Execute the collision step locally with f_i by Eq. (16)

Streaming

Denote the interpolation as $H(f)$
for each $i \in [9,0)$ **do**
  $id = 10 \times (\eta \times NX_{\xi} + \xi)$
  $f[id + (i+t+1)\%10] = H(f^{*}[id + (i+t)\%10])$
end for

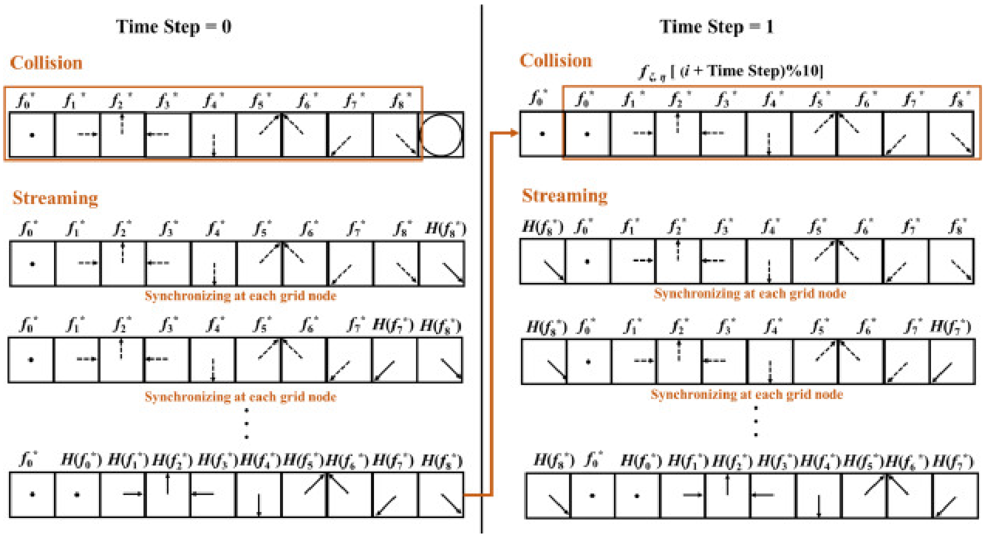

Fig. 1. Schematic diagram of the present index algorithm for lattice node (ξ, η), where the memory address in Eq. (22) is denoted as $f_{\xi,\eta}[i]$.

In the original two-lattice algorithm for $DdQq$ scheme, $2q$ distribution functions are needed, but the present algorithm can reduce the number of distribution functions to $(q+1)$. As shown in Table 1, together with the macroscopic variables, the two-lattice algorithm for the GICLBM utilizes a memory array with a size of $(d \times q + 2q + d + 1) \times N_{node}$, whereas the present algorithm reduces this value to $(d \times q + q + d + 2) \times N_{node}$. Moreover, although the GICLBM is described in a two-dimensional form, this low-memory usage streaming algorithm is described as the $DdQq$ scheme and can be extended to the three-dimensional situations in a straightforward manner.

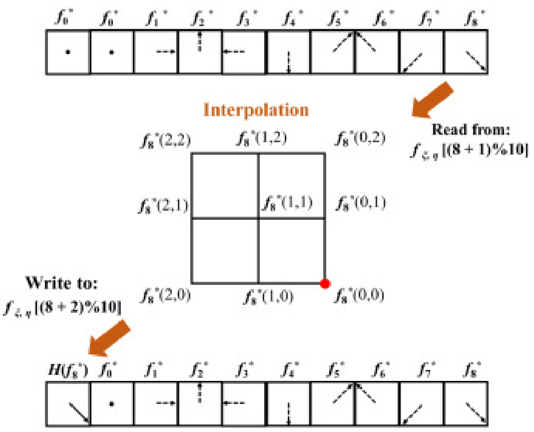

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Fig. 2. Schematic diagram of the streaming step of $f_{\xi,\eta}(8)$ as an instance, when the time step is equal to 1.

The solver based on the present GICLBM is programmed by CUDA C. All cases presented in this work are executed on a CPU processor of Intel Xeon E5-2678v3 and accelerated using a single graphic processing unit (GPU) of NVIDIA TITAN V. The whole computational time, including the pre-processing and the post-processing, is provided at the end of each case.

Table 1. Comparisons of the array size between the present and the two-lattice algorithm for GICLBM, where $N_{node} = NX_{\xi} \times NY_{\eta}$.

Array name	Present algorithm	Two-lattice algorithm
Particle distribution function f	$(q+1) \times N_{node}$	$2q \times N_{node}$
Density ρ	N_{node}	N_{node}
Velocity u	$d \times N_{node}$	$d \times N_{node}$
Streaming distance $\Delta\xi$	$d \times q \times N_{node}$	$d \times q \times N_{node}$
Overall	$(d \times q + q + d + 2) \times N_{node}$	$(d \times q + 2q + d + 1) \times N_{node}$

3. Results and discussions

3.1. Case I: A square cylinder

The first case considers the flow past and noise radiated from a square cylinder with uniform upstream flow. The flow pattern varies with the Reynolds number (Re), defined as $Re = U_0 L_0 / \nu$. Here the characteristic length L_0 is set as the edge length of the square. When Re exceeds the critical value, alternating vortex shedding appears in the downstream wake. The physical phenomenon is similar to the well-known flow past a circular cylinder. A key difference between these two types of flow lies in the location of the separation point. The separation point of a circular cylinder is dynamically varying within a certain range, whereas for the square cylinder, the separation occurs at either the windward corner of the square ($Re > 120$) or the leeward corner ($Re < 120$) [41], [42].

In the present work, we focus on the noise generated from the surface of the square cylinder at $Re = 150$, when the flow field is in the laminar regime, and the two-dimension assumption can be satisfied. As depicted in Fig. 3(a), the origin of the Cartesian coordinate is located at the center of the square, and the x direction is parallel to the direction of U_0. Moreover, the polar coordinate is also set in the center of the square. The axis of $\theta = 0^\circ$ coincides with the positive direction of the x axis, and the counterclockwise is the positive direction. The O-type mesh is utilized in this case. As shown in Fig. 3(b), the mesh along the square surface is gradually refined from the midpoint to sharp corners, and stretched along the radial direction. The radius of the physical domain is close to $200L_0$ and the width of the non-reflective buffer layer is close to $80L_0$ which is larger than the wavelength of sound waves.

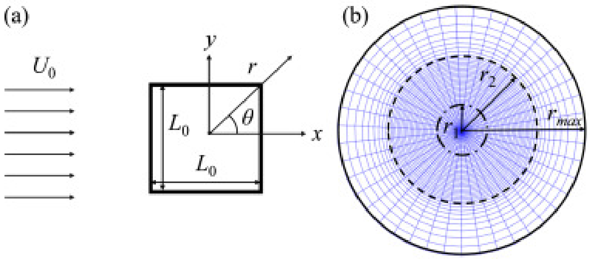

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Fig. 3. (a) Schematic diagram of flow past a square cylinder and (b) the body-fitted mesh in the physical domain.

For ensuring the accuracy of fluid dynamics and the far-field noise, the physical domain is divided into three regions. The first region is near the square cylinder with various nonlinear fluid structures. The mesh spacing varies exponentially from the first mesh spacing, which is denoted by Δr_0. The second region is that sound waves linearly propagate, and the mesh spacing remains consistent. The third region is the non-reflective buffer layer where the sound waves dissipate. The mesh is extended exponentially to a relatively coarse resolution. The variation of the mesh spacing can be expressed as

$$\begin{aligned}
&\text{Region I (exponential)}: \Delta r_\eta = \Delta r_0 e^{\gamma \frac{\eta}{NY_{\eta 1}}}, 0 \le \eta \le NY_{\eta 1},\\
&\text{Region II (} \quad \text{linear} \quad \text{)}: \Delta r_\eta = \Delta r_{NY_{\eta 1}}, NY_{\eta 1} < \eta < NY_{\eta 2},\\
&\text{Region III (exponential)}: \Delta r_\eta = \Delta r_0 e^{\gamma \frac{\eta - NY_{\eta 2} + NY_{\eta 1}}{NY_{\eta 1}}}, NY_{\eta 2} \le \eta < NY_\eta,
\end{aligned} \tag{23}$$

where $NY_{\eta 1}$ and $NY_{\eta 2}$ are the interface of different regions in the computational domain, and the variation of the mesh spacing can be adjusted by γ.

The Strouhal number (St) is defined as $St = f_v L_0 / U_0$, where f_v is calculated within an interval of more than 30 periods using the fast Fourier transform (FFT). The force coefficients are normalized by

$$C_D = \frac{F_x}{\frac{1}{2}\rho_0 U_0^2 L_0}, \quad C_L = \frac{F_y}{\frac{1}{2}\rho_0 U_0^2 L_0}, \tag{24}$$

where C_D and C_L are the drag and lift coefficients, respectively. Both F_x and F_y are the force components acting on the square cylinder, and can be obtained via integrating the shear stress and pressure tensor along the surface.

The mesh convergence study and validation are carried out in Table 2, where the CFL number is fixed as 0.9. Three mesh configurations with different $NX_\xi \times NY_\eta$ are equal to 256 × 384, 512 × 768 and 1024 × 1152, which is corresponding to Δr_0 are 0.01, 0.005 and 0.003, respectively. $[]_{rms}$ denotes the root-mean-square (rms) value which is calculated over 30 vortex shedding periods. $[]'$ is the maximum of the fluctuation. In Table 2, the maximum difference between the coarsest and finest mesh is larger than 5% in C_L'. The results from the middle mesh resolution are approaching that of the finest mesh. As a result, it confirms that the middle mesh is sufficient for simulating the fluid dynamic and noise generated from the square cylinder in the laminar regime.

As shown in Fig. 4(a), the positive and negative vortex can alternately shed from the square cylinder and move downstream at $Re = 150$. The vorticity is defined as $\omega = 0.5 \times (\partial_x u_y - \partial_y u_x)$, which can be calculated in the computational domain and transformed to the physical domain. In Fig. 4(a), the lines are identified by $\lambda_{ci} = 0.2$ [46]. The solid and dash lines correspond to the positive and negative ω, respectively.

Table 2. Mesh convergence study and validation for flow past a square cylinder. Here p'_{rms} is measured at $r = 80L_0$ and $\theta = 90^\circ$.

Re	Case	No. of mesh cells $(NX_\xi \times NY_\eta)$	St	$\bar{C_D}$	C_L'	$C_{L,rms}$	p'_{rms} $(\times 10^{-5})$
100	Present	**256 × 384**	0.138	1.436	0.271	0.191	3.615
		512 × 768	0.145	1.448	0.257	0.182	3.521
		1024 × 1152	0.146	1.462	0.257	0.180	3.528
	Sohankar et al. [43]	–	0.146	1.477	–	–	–
	Jiang and Cheng [44]	–	0.143	1.443	0.251	–	–
	Bai and Alam [45]	–	0.146	1.48	–	–	–
	Kumar et al. [3]	–	0.1407	1.4403	0.2499	0.1767	–
150	Present	**512 × 768**	0.1530	1.418	0.389	0.275	5.360
	Komatsu et al. [2]	–	0.1527	1.449	–	0.2676	5.239

Under the laminar condition at $Re = 150$, as shown in Fig. 4(b), the instantaneous C_L and C_D periodically vary due to the vortex shedding. Phases of the present curves are shifted to align the first moment where C_L varies from positive to negative values. In Fig. 4(b), it is also shown that the present results are in good agreement with Komatsu et al. [2]. The frequency spectrum is depicted in Fig. 4(c), the frequency of C_D is 0.306 and double to C_L, 0.153. The time-averaged pressure C_p and the skin-friction coefficient C_f coefficient are calculated by

$$C_p = \frac{\bar{p}}{\frac{1}{2}\rho_0 U_0^2}, \quad C_f = \frac{\bar{F}_f}{\frac{1}{2}\rho_0 U_0^2}. \tag{25}$$

The distribution of C_p and C_f are symmetric about the center line, $y = 0$. As shown in Figs. 5(a) and 5(b), it can be clearly found that the separation point is located at the sharp corner of the windward side in $Re = 150$. Both C_p and C_f dramatically change at the sharp corner because of the intense velocity gradient. The time-averaged pressure C_p calculated by the coarsest mesh oscillates at the sharp corner, and deviates from the finest mesh. It can also be found that 512 × 768 with $\Delta r_0 = 0.005$ is already efficient enough for the laminar regime.

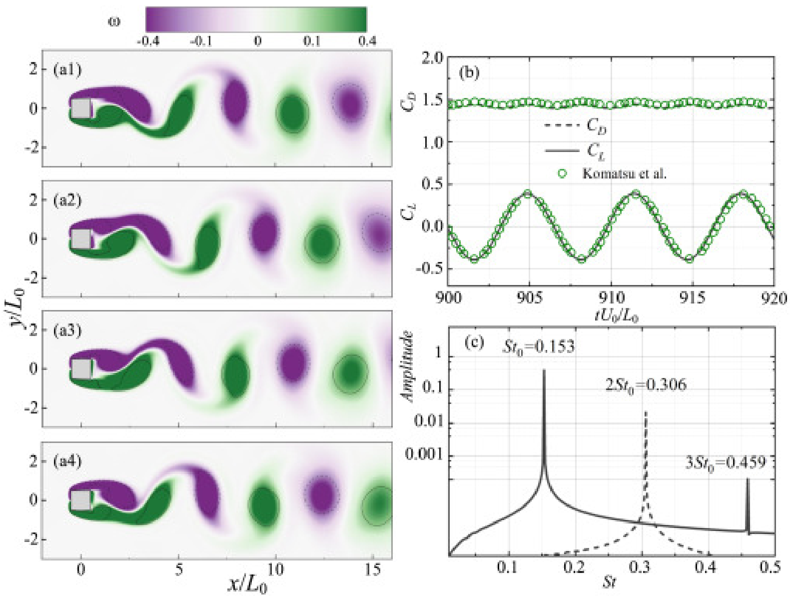

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Fig. 4. At $Re = 150$ and $Ma = 0.2$, (a) instantaneous vorticity contours, (b) time histories of the lift coefficient C_L and the drag coefficient C_D, and (c) the frequency spectrum of force coefficients through FFT. Results are compared with those of Ref. [2], which are denoted by green circles.

The wavelength of the sound wave can be estimated from the period of the vortex shedding. In order to avoid the errors of dispersion and dissipation in the sound propagation, more than 60 mesh points are settled in one single sound wave even for the coarsest mesh. This mesh resolution is much higher than 6 meshes generally required for the LBM in acoustic problems [5], [6], [47].

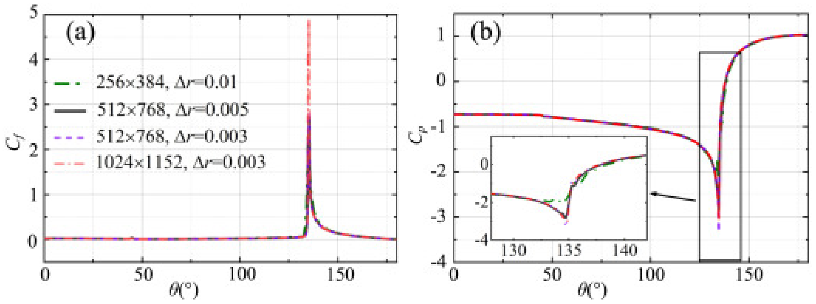

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Fig. 5. At $Re = 150$ and $Ma = 0.2$, (a) the skin-friction coefficient C_f along the surface and (b) the time-averaged pressure coefficient C_p.

The dilatation is defined as $\dot{p} = \partial p/\partial t$, and can be used to describe the noise radiation in the far-field. Similar to the noise radiated from the circular cylinder, positive and negative pressure pulses are periodically generated from the surface of the square cylinder and propagate outward, as shown in Fig. 6.

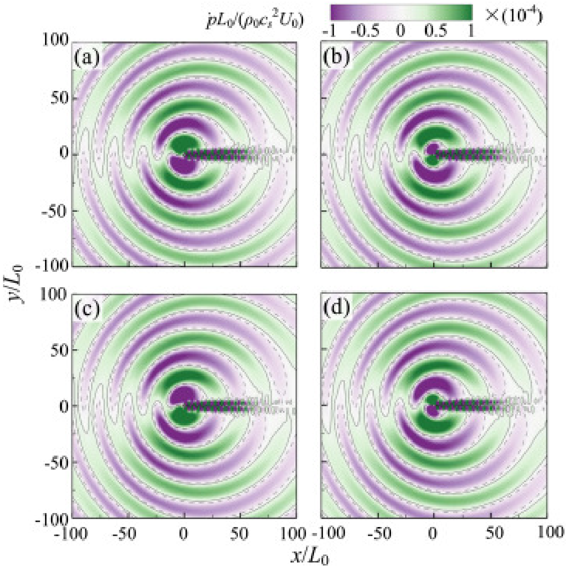

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Fig. 6. The noise radiated from the square cylinder represented by the dilatation at $Re = 150$ and $Ma = 0.2$. The solid and dashed isolines denote the value of ± 0.00001, respectively.

The instantaneous fluctuating pressure is defined as $p' = p - \bar{p}$. And the time history of p' measured at $(0, 80L_0)$ also changes periodically. Its frequency is basically consistent with that of C_L, as shown in Fig. 7(b). Arranging the observer points along $\theta = 90^\circ$, the decay of sound waves is proportional to $1/\sqrt{r}$, according to the two-dimensional Curle's solution [4]. The acoustic radiation directivity is calculated by p'_{rms} and depicted in the polar coordinate. The observer points are located at $r = 80L_0$ with an equal-size angle interval. The far-field noise radiation is dominated by the lift dipole. And the acoustic results are in good agreement with those of Ref. [2], [3].

Simulating the square cylinder with the mesh configuration of 512 × 768 and $\Delta r_0 = 0.005$ takes 2.5 h to compute 600 times the characteristic time which is defined as L_0/U_0.

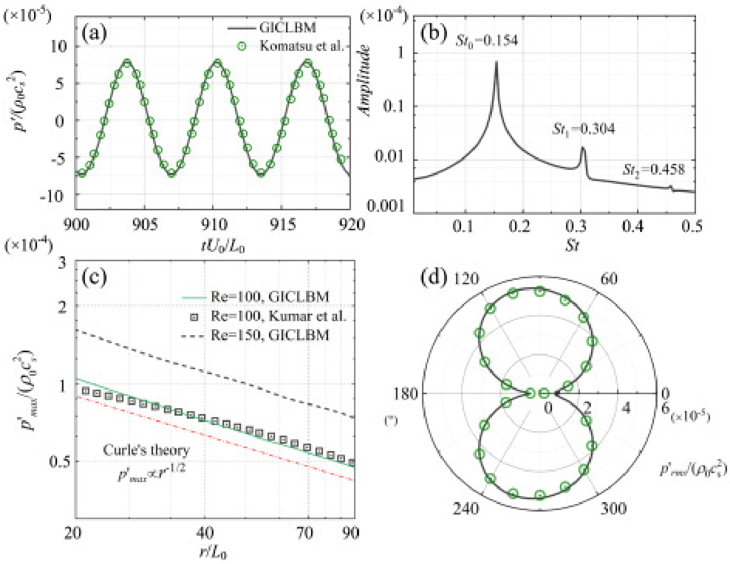

Fig. 7. At $Re = 150$ and $Ma = 0.2$, (a) the time history of the pressure fluctuation p' measured at $(0, 75L_0)$ and (b) its frequency spectrum. (c) The decay of the maximum pressure fluctuation along $\theta = 90^\circ$, where the green solid and black dashed lines are the present results at $Re = 100$ and $Re = 150$, respectively. The red dash–dot line is the theoretical decay trend, $\propto 1/\sqrt{r}$, and black squares are results from Ref. [3]. (d) The acoustic radiation directivity is represented by p'_{rms} and measured at $r = 80L_0$. Green circles denote results of Ref. [2].

3.2. Case II: An elliptic cylinder

The noise radiated from the flow past an elliptic cylinder is investigated in this section. Similar to the square cylinder, when Re exceeds the citric value, the incoming flow will separate from the surface and occur the periodically vortex shedding. Moreover, the fluid dynamics and noise generated from the ellipse are affected by both the aspect ratio (AR) and the angle of attack, α. As shown in Fig. 8, the aspect ratio AR is defined as the ratio of the minor axis to the major axis. The Reynolds number Re is calculated with the major axis as the characteristic length.

Referring to the work of Mittal and Balachandar [48], the O-type elliptic cylindrical mesh is generated and briefly described here. The mesh in the physical domain can be obtained by the transformation from the elliptic coordinate (EX_ξ, EY_η) to the Cartesian coordinate (x, y), which is calculated by

$$x = h \cosh EX_\xi \cos EY_\eta, \quad y = h \sinh EX_\xi \sin EY_\eta, \tag{26}$$

where h is the distance from the center of the ellipse to the focus. In the present work, the ellipse boundary is described as

$$EX_0 = \tfrac{1}{2}\ln[(1 + AR)/(1 - AR)], \quad EY_\eta = \tfrac{2\pi(\eta)}{N_\theta}, \quad \eta = 0, 1, \ldots, N_\theta. \tag{27}$$

The distance h is calculated by $h = 1/\cosh E_{\xi,0}$. The mesh is extended to the outer boundary following

$$EX_\xi = \ln\left[r(\xi)\cosh EX_0 + \sqrt{r(\xi)^2 \cosh^2 EX_0 - 1}\right], \tag{28}$$

where r is the distance between the center and the outer ellipse. Similar to the setup of the square cylinder, the physical domain is divided into three regions by controlling the variation of Δr. Moreover, as r increased, the outer boundary of the physical domain approaches a circle. The radius is close to $200L_0$ and the thickness of the non-reflective buffer layer is close to $80L_0$.

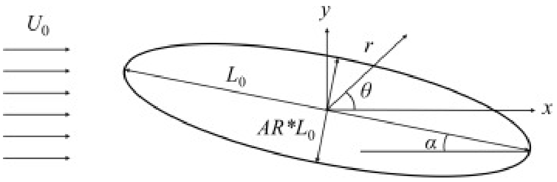

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Fig. 8. Schematic diagram of flow past an elliptic cylinder with an angle of attack α.

The mesh configuration of 512 × 768 with $\Delta r_0 = 0.005$ is also selected for this case. At $Re = 150$ and $\alpha = 90^{\circ}$, results obtained by GICLBM are validated by comparing with Ref. [49], [50]. It should be noted that $[]^*$ denotes that the projected length is the characteristic length. In Table 3, the largest relative deviation is 4.2%, corresponding to $\bar{C_D}$ at $AR = 0.75$.

Variations of wakes behind the elliptic cylinder with the different α are depicted in Fig. 9. The downstream wakes are basically the same, but the spacing of the adjacent vortex decreases with the increase of α. This phenomenon indicates that the frequency of vortex shedding is also increasing. Three wake patterns have been reported by Shi et al. [49] about the flow past the elliptic cylinder on various combinations of AR and α. The first region is the steady wake as shown in Fig. 10(a), where the vortex shedding is in the absence. The second region is the Kármán vortex followed by the steady wake as shown Fig. 9, where the vortex can alternately shed from the cylinder and develop to a steady flow in the downstream wake. As for the third region, the steady wake in the second region is in instability again and will develop into another Kármán vortex at a different frequency compared to that of the upstream vortex shedding. As reported before, the third region occurs when $A < 0.67$ and $\alpha > 52^{\circ}$ [49], but is not found in the present investigation. This is probably due to the bias of selection in Re rather Re^*.

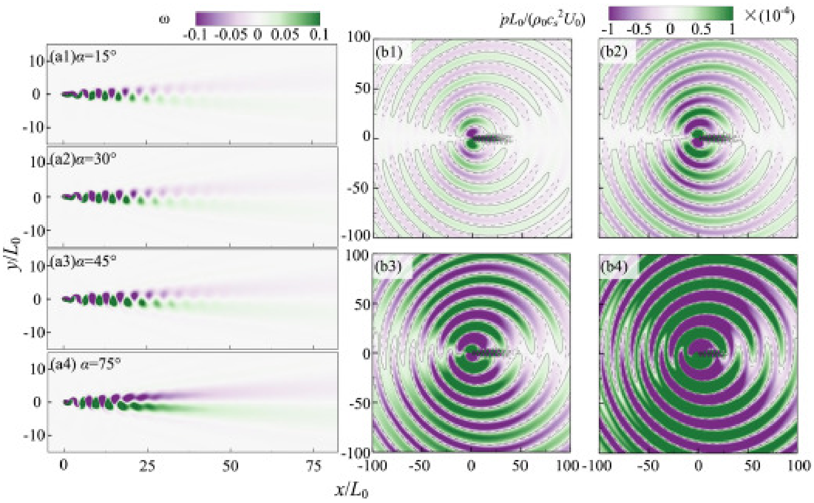

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Fig. 9. The variation of (a) vorticity and (b) noise radiation represented by the dilatation with the angle of attack α at $Re = 150$, $Ma = 0.2$ and $AR = 0.6$.

Correspondingly, the wavelength of the sound wave radiated from the elliptic cylinder for different AR is also enlarged. As α increases, the intensity of noise is enhanced and the sound waves are also gradually appearing in the direction of the drag.

In addition, we also compare the effects of AR on wake patterns and fluid dynamics at $\alpha = 4^{\circ}$. The instantaneous vorticity contours are depicted in Fig. 10. When $AR = 0.2$, no vortex shedding appears which might be because Re calculated by the major axis is lower than the critical value for vortex shedding. In other AR, the distance between two adjacent vortices increases with the increase of AR.

Table 3. Comparisons of force coefficients for the elliptic cylinder with $\alpha = 90^\circ$ at $Re = 150$.

	Present		**Shi et al. [49]**		**Pradhan et al. [50]**	
AR	$\bar{C_D}^*$	St^*	$\bar{C_D}^*$	St^*	$\bar{C_D}^*$	St^*
0.25	2.059	0.164	2.068	0.167	2.153	0.17
0.5	1.848	0.191	1.824	0.191	1.8674	0.19333
0.75	1.62	0.195	1.555	0.192	1.5907	0.19333

The trend of forces with varying AR is shown in Fig. 11 and compared with that in the study of Mahato et al. [51]. Both the mean C_D and the rms value of C_L increases with AR, because the projected length of the elliptic cylinder becomes larger. Moreover, the shape of the ellipse becomes close to the circle as AR increases. The mean C_L is slightly affected by α, and tends to be zero as $AR \approx 1$. In Fig. 11(d), St decreases with AR, as evidenced by the increase of the distance between two adjacent vortices in Fig. 10.

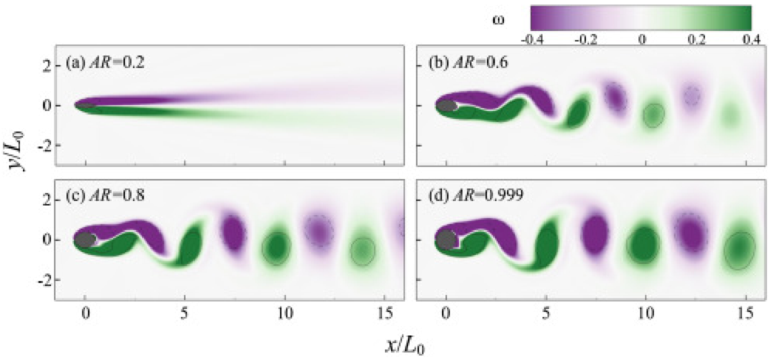

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Fig. 10. Instantaneous vorticity contours with different ARs at $Re = 150$.

At $AR = 0.2$, the noise level is quite low, because vortex shedding does not occur in this condition and the flow is steady. It can be seen in Fig. 12 that the intensity of noise increases with AR, which is similar to the trend of the vorticity strength. Moreover, the noise radiation is gradually enhanced along the x direction. Similar to the above results of flow past a square cylinder, it is confirmed that the frequency of the sound wave and the vortex shedding are identical. The wavelength of the sound wave also decreases with AR, which is consistent with the trend of St in Fig. 11(d). The p'_{rms} measured by the observer point at $r = 75L_0$ is slightly lower than that of Ref. [51].

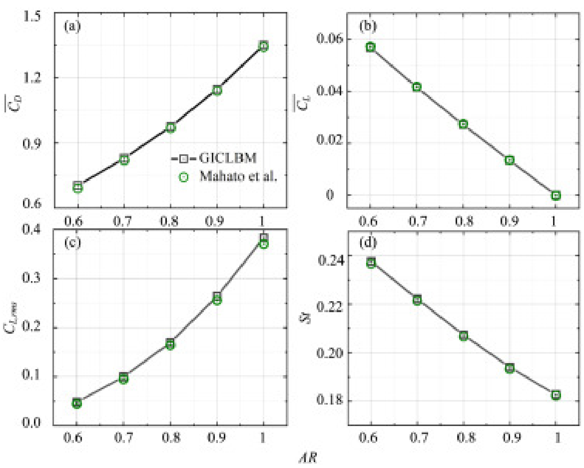

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Fig. 11. The variation of fluid dynamics with AR at $Re = 150$ and $Ma = 0.2$. These coefficients, including (a) the mean drag, (b) the mean lift, (c) the root-mean-square value of the lift, and (d) the Strouhal number, are statistically calculated in an interval of more than 30 vortex shedding periods.

Furthermore, although the first mesh spacing is controlled to be the same as both the square and the elliptic cylinders, the contravariant velocities $\widetilde{e_i}$ are different because of the distinct shapes of cells adjacent to the wall. And corresponding time steps for calculating the elliptic cylinder will be larger than that of the square cylinder. About 3.3 h is spent for simulating flow past the elliptic cylinder with 600 times the characteristic time, be slower than flow past the square cylinder. Results in this section prove that the GICLBM can accurately simulate the unsteady flow and noise radiated from an elliptic cylinder in the laminar condition with the O-type mesh setup.

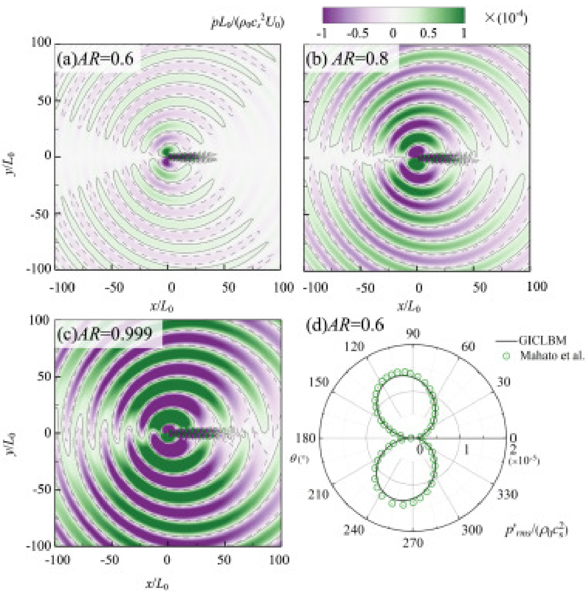

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Fig. 12. Noise radiation represented by the dilatation when AR is (a) 0.6, (b) 0.8, and (c) 0.999 at $Re = 150$ and $Ma = 0.2$. The solid and dashed isolines denote the value of ± 0.00001, respectively. (d) The acoustic radiation directivity is represented by p'_{rms} measured at $r = 75L_0$ where the green circles denote results of Ref. [51].

3.3. Case III: A NACA 0012 airfoil

In this section, we introduce the third case, i.e., flow past a NACA 0012 airfoil and the associated noise radiation. The airfoil is immersed in a uniform incoming flow at an angle of attack of α, and the vortex will shed from the suction side. The noise is mainly generated from the trailing edge of the airfoil and radiated to the far-field. As shown in Fig. 13, the origins of polar and Cartesian coordinates are both located at the leading edge of the airfoil. The chord length L_0 is set as the characteristic length. The C-type mesh is generated with the tail that has an uncut and sharp edge. Similarly, as shown in Fig. 13, the mesh spacing variation in the physical domain is divided into three regions according to the distance away from the airfoil and same as the above two cases. In addition, when $x < 1$, the mesh is stretched from the airfoil to the outer boundary of the physical domain. The mesh behind the airfoil is also stretched from $x = 1$ and $y = 0$, respectively. The size of the physical domain, r_{max} is determined to be close to $50L_0$, and the width of non-reflective sound boundary layer is close to $10L_0$.

At $Ma = 0.2$, two typical two-dimensional cases, $Re = 1000$ and $Re = 5000$, are investigated with $\alpha = 10^\circ$. Similar to the ellipse cylinder with an angle of attack, the shear layer separates from the suction side. Firstly, three mesh configurations are implemented for the validation. The mesh configurations include $NX_\xi \times NY_\eta$, which are 1151 × 384, 2303 × 768 and 2303 × 768, respectively. Correspondingly, we have Δr_0 being 0.005, 0.001 and 0.0002, respectively. Due to the CFL number being fixed at 0.9, the time consumption of the third mesh is 5 times longer than the second mesh. The results are summarized in Table 4. It is found that Δr_0 has significant effects on the fluid dynamic. At $Re = 5000$, the relative deviation of St between first and second meshes is larger than 60%.

Coarsest meshes will lead to erroneous results. The result of second mesh is in the best agreement with the references, and the maximum relative error is less than 4%. When Δr_0 continues to be refined, the relative error increases. Therefore, the second mesh is used to conduct the direct acoustic simulation.

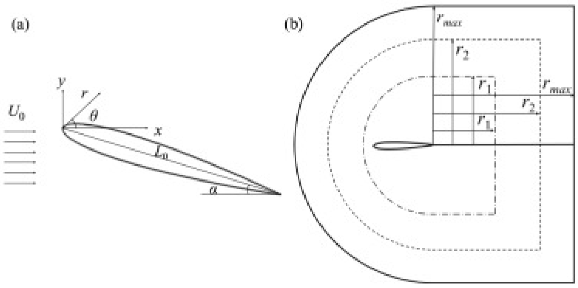

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Fig. 13. (a) Schematic diagram of flow past a NACA 0012 airfoil and (b) the C-type mesh configuration.

As shown in Fig. 14 at $Re = 1000$, the alternate vortex shedding appears downstream. The time history of the lift coefficient is close to a sinusoidal curve. The positive and negative pressure pulses are generated from the trailing edge as the vortex sheds, and propagate to the far-field.

Table 4. Comparisons of force coefficients and vortex shedding frequency for flow past a NACA 0012 airfoil at $\alpha = 10^\circ$ and $Ma = 0.2$.

Re	Case	No. of mesh cells $(NX_\xi \times NY_\eta)$, Δr_0	St	$\bar{C}_D$	$\bar{C}_L$	$C_{L,max} - C_{L,min}$
5000	Present	1151×384, $\Delta r_0 = 0.005$	1.100	0.148	0.525	0.401
		2303×768, $\Delta r_0 = 0.001$	0.661	0.187	0.960	0.606
		2303×768, $\Delta r_0 = 0.0002$	0.658	0.187	0.974	0.605
	Hatakeyama and Inoue [52]	–	0.66	0.183	0.937	–
	Falagkaris et al. [53]	–	0.683	–	–	–
1000	Present	2303×768, $\Delta r_0 = 0.001$	0.870	0.167	0.420	0.055
	Wang and Tian [54]	–	–	0.172	0.419	0.055
	Fang et al. [55]	–	–	0.167	0.420	0.051
	Falagkaris et al. [53]	–	0.861	–	–	–
	Johnson and Tezduyar [56]	–	0.86	–	–	–

At $Re = 5000$ and $\alpha = 10^\circ$, the instantaneous vorticity contours are depicted in Fig. 15. Flow separates from the suction side and generates a laminar separation bubble.

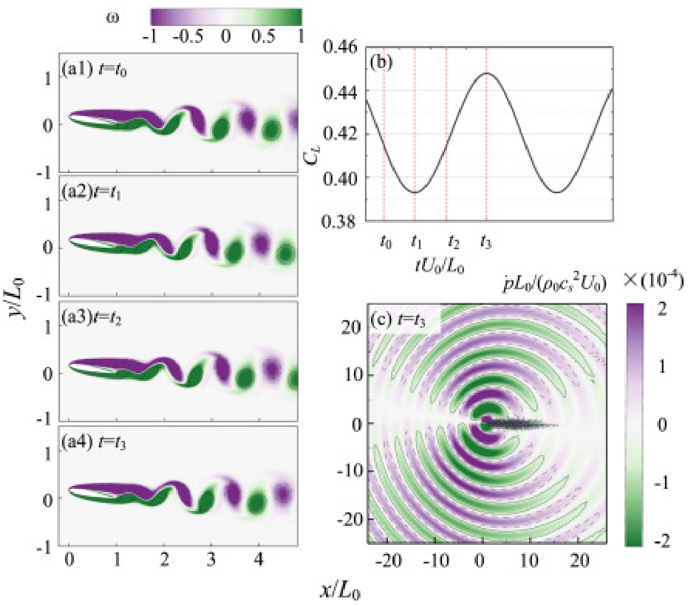

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Fig. 14. (a) Instantaneous vorticity snapshots at $\alpha = 10^\circ$, $Re = 1000$ and $Ma = 0.2$ at four typical time instants, t_0, t_1, t_2 and t_3, as shown in (b). (c) Noise radiation at $t = t_3$ is represented by the sound dilatation, where solid and dashed isolines denote the value of ± 0.001, respectively.

As shown in Fig. 16, the vortex shedding from the suction side makes the forces exerting on the airfoil change periodically and the phase-diagram of $C_L - C_D$ exhibits a clear and closed curve. Utilizing the FFT to obtain the frequency spectrum of C_L and C_D, it can be found that both the fundamental and harmonic frequencies coincide, which is different from the flow past a square cylinder. The distribution of the pressure coefficient along the airfoil surface is shown in Fig. 16(d), which is in good agreement with that of Hatakeyama and Inoue [52] using the finite difference method.

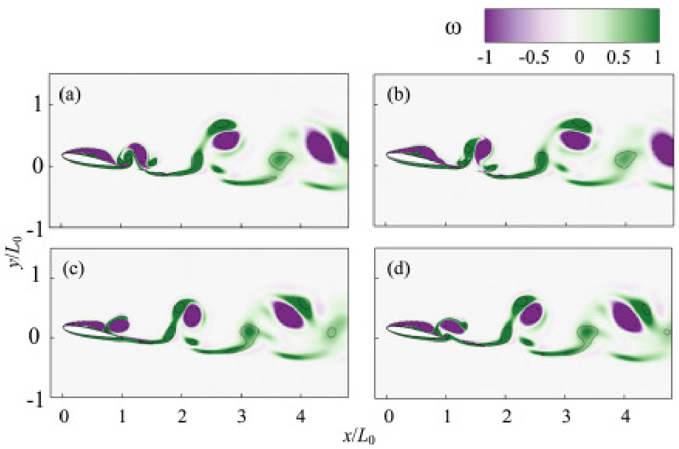

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Fig. 15. Instantaneous vorticity contours at four typical time instants at $Re = 5000$ and $\alpha = 10^\circ$.

The noise radiated from the flow past the airfoil is similar to the square and elliptic cylinders. The pressure pulses are generated from the trailing edge due to the alternately vortex shedding and propagate to the far-field. The sound waves radiated to the positive and negative directions of the y axis are not consistent because of the existence of α, and different wavelengths of sound waves occur in the far-field (see Fig. 17).

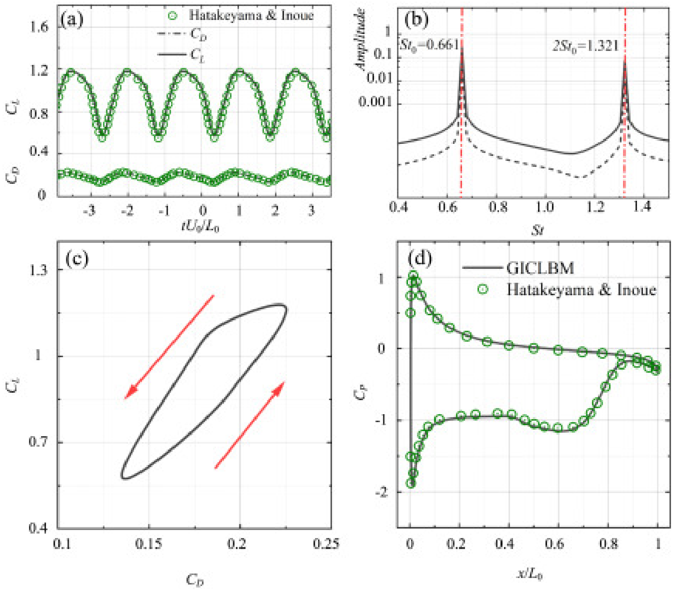

Fig. 16. (a) The time history, (b) the frequency spectrum, and (c) the phase diagram of the lift and drag coefficients at $Re = 5000$, $Ma = 0.2$, and $\alpha = 10^\circ$. (d) The distribution of the time-averaged pressure coefficient C_p. Green circles denote results of Ref. [52].

The time history of $p - p_0$ is measured with the observer located at $r = 30$, which is depicted in Figs. 18(a) and 18(b). It should be noted that the present results are not consistent with Hatakeyama and Inoue [52], possibly due to the difference in the background pressure p_0. For ease of comparison, the present curves are shifted to ensure an identical maximum value. In Figs. 18(a) and 18(b), it can be seen that the present results are in good agreement with Hatakeyama and Inoue [52]. The curves measured at $\theta = 90^\circ$ and $\theta = 270^\circ$ are no longer symmetrical with the x axis. As shown in Fig. 18(c), the frequency spectrum of p' is composed of the fundamental and harmonic tones at $St = 0.661$ and $St = 1.3211$, which is similar to the frequency spectrum of C_L. In other words, under the condition of $Re = 5000$, $Ma = 0.2$ and $\alpha = 10^\circ$, the consistency in frequency spectrum of C_L and the pressure fluctuation indicates that the aeolian tones are generated from the periodical vortex shedding. This is different from the widely accepted acoustic feedback mechanism of the noise generated from the tailing edge. The significant phenomenon is that some multiple tones occur around a main tone in the frequency spectrum [57], [58]. It is believed that Tollmien–Schlichting (T–S) waves from the tailing edge are enhanced by laminar separation bubbles, and induce the instability at the leading edge. Therefore, a self-sustaining acoustic feedback loop appears. However, it is still controversial whether the laminar separation bubble on the suction side or the pressure side plays the main role. This disappearance may be because α chosen in this paper is relatively large, and the vortex shedding is still the dominating mechanism of noise generation. Similar results have been reported in references within a higher α range [52], [54]. The noise radiation in the far-field is dominated by the lift dipole, which is consist with Ref. [52]. The current validations confirm the accuracy of GICLBM within the investigated parameter range. However, it is important to note that our study does not verify the capability of GICLBM to handle problems beyond the weakly compressible constraint.

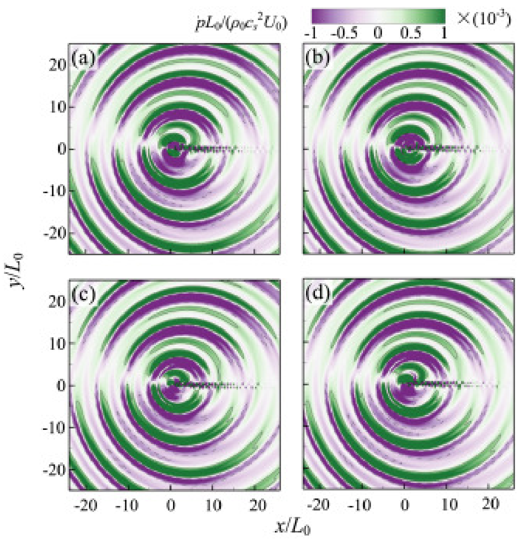

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Fig. 17. Noise radiated from the NACA 0012 airfoil represented by the dilatation at four typical time instants at $\alpha = 10^\circ$, $Re = 5000$, and $Ma = 0.2$. The solid and dashed isolines denote the value of ± 0.001, respectively.

Thanks to the reduction of the vortex shedding period, only 150 times the characteristic time is enough to obtain reliable results. The computation on a mesh of 2303 × 768 and $\Delta r_0 = 0.001$ takes 11.8 h, which is much more expensive than simulating flow past the square or the elliptic cylinder.

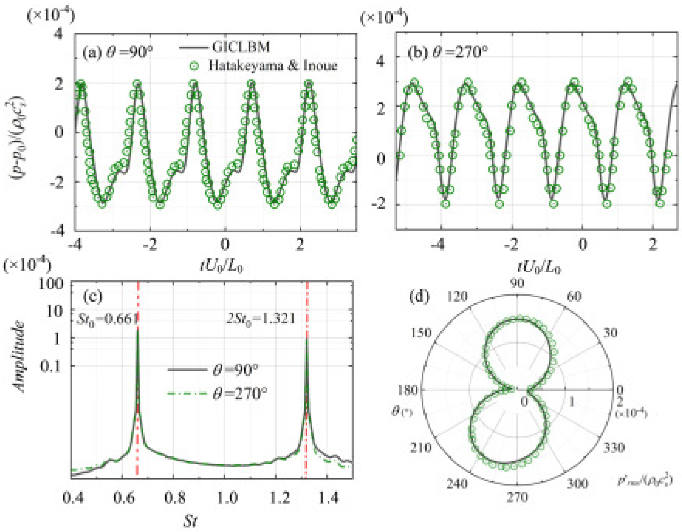

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Fig. 18. Time history of $p - p_0$ measured at (a) $\theta = 90^\circ$ and (b) $\theta = 270^\circ$ with $r' = 30L_0$ and (c) their frequency spectrum when $Re = 5000$, $Ma = 0.2$ and $\alpha = 10^\circ$. (d) The acoustic radiation directivity represented by p'_{rms} measured at $r = 30L_0$. Green circles denote results of Ref. [52].

4. Conclusion

The improved lattice Boltzmann method that includes the cascaded collision and generalized interpolation-supplemented streaming is introduced and discussed with three typical external flow problems. With this method, non-uniform and body-fitted mesh partitions are allowed, tackling the vital problem of conventional lattice Boltzmann method that is usually limited to the uniform mesh. By incorporating the perfectly matched layer that avoids sound waves being reflected to the inner computational zone, direct simulations of noise radiation in external flows become possible.

By conducting cases including flow past a square cylinder, flow past an elliptic cylinder, and flow past a NACA 0012 airfoil, the accuracy of present GICLBM is investigated. Results related to fluid dynamics such as lift and drag, vortex shedding frequency, wall pressure distribution, etc., agree well with the reference data based on either lattice Boltzmann or Navier–Stokes solvers. Coarse meshes can yield barely satisfactory results for square and elliptic cylinders in all Reynolds number conditions discussed in this work. However, for the NACA 0012 airfoil in larger Reynolds numbers with a 10° angle of attack, the mesh with a relatively coarse first mesh spacing is unable to sufficiently resolve the flow dynamics and provide accurate results. Within parameters considered in the present study, the aeoline tones in far-field are dominant and indicate a direct relationship with the vortex shedding. For all cases discussed in this study, good accuracy is realized in terms of sound radiation directivity, sound intensity, and the decaying law of noise amplitude in the far-field.

In addition, the memory-saving streaming algorithm is proposed for the GICLBM. By recalculating the index based on the lattice node position, the current time step and the discrete velocity's direction, the memory usage is reduced significantly.

In summary, case studies presented here confirm that the GICLBM is feasible for CAA problems at low Mach numbers, with balanced computational accuracy and efficiency. In the future, the present study will be extended to solve turbulent flow problems, so as to provide an easy-to-use, reliable, and efficient tool for scientific research and engineering applications.

CRediT authorship contribution statement

Jian Song: Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Fan Zhang:** Visualization, Formal analysis, Data curation. **Yuanpu Zhao:** Visualization, Formal analysis, Data curation. **Feng Ren:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Haibao Hu:** Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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